SEAH ZHANG JIAN

A24CS0297

Lab 2: Biological Sequences as Strings

Clinical genomics sequence screening: build the first functions of your sequence toolkit

Item	Detail
Mode	Individual
Weight	2% of the course mark
Teaching week	Week 2 (19 - 25 Oct 2026)
CLOs	CLO1
Competencies	C01, C05, C11
AI level	Level 1: AI as tutor (PROPOSED policy). You may ask AI to explain concepts, give hints, explain an error message, or review code you wrote. You may NOT paste in the task and submit AI-written code or text as your answer.
AI log	Submit AI_USE_LOG.md (prompt log), or a signed nil return if you used no AI.

Overview

A clinical genomics lab checks short DNA sequences before they go into an assay. In this lab you write five small, tested functions that clean, validate and analyse a sequence. In Week 5 these functions become your sequence_utils.py module, so write them carefully now.

Work in the starter repository. Every function has a name, a signature and a docstring. Replace the line `raise NotImplementedError` with your own code. Do not change the function names or signatures.

What you will practise

- Index, slice and transform strings; call string methods; build strings with join and f-strings.
- Validate input before using it and raise ValueError with a clear message.
- Handle boundary cases: empty string, lower case, whitespace, the ambiguous base N.
- Write and run pytest tests; keep the code clean enough to pass ruff.

Files and resources

- Starter repository: src/lab02_sequences.py (five stubs), tests/test_visible.py (visible tests), pytest.ini, requirements.txt, AI_USE_LOG.md.
- Run the tests with: python -m pytest. Check style with: ruff check src.

Instructions

- Create a virtual environment and install requirements.txt (see the Week 1 notes).
- Complete the four function tasks in src/lab02_sequences.py. Commit after each task with a clear message.
- Add at least three tests of your own for find_motif in tests/test_lab02_sequences.py (one normal, one boundary, one invalid input).
- Complete AI_USE_LOG.md. This lab is AI Level 1 (tutor): AI may explain and give hints but must not write your answers.

Tasks

T1. clean_sequence and is_valid_dna (3 marks)

clean_sequence(raw) upper-cases raw text and keeps letters only (drops spaces, digits, new lines). is_valid_dna(seq, allow_n=False) is True only for a non-empty string of A, C, G, T (and N when allow_n is True). The empty string is NOT valid DNA.

Hint: set(seq) <= set('ACGT') is True for an empty string, so test for emptiness first.

T2. gc_content (3 marks)

gc_content(seq) returns the percentage (0 to 100) of G and C. Raise ValueError for an empty or invalid sequence. Lower case must work.

T3. transcribe (2 marks)

transcribe(dna) returns the RNA copy (T becomes U), in upper case. Raise ValueError for invalid DNA.

T4. find_motif (4 marks)

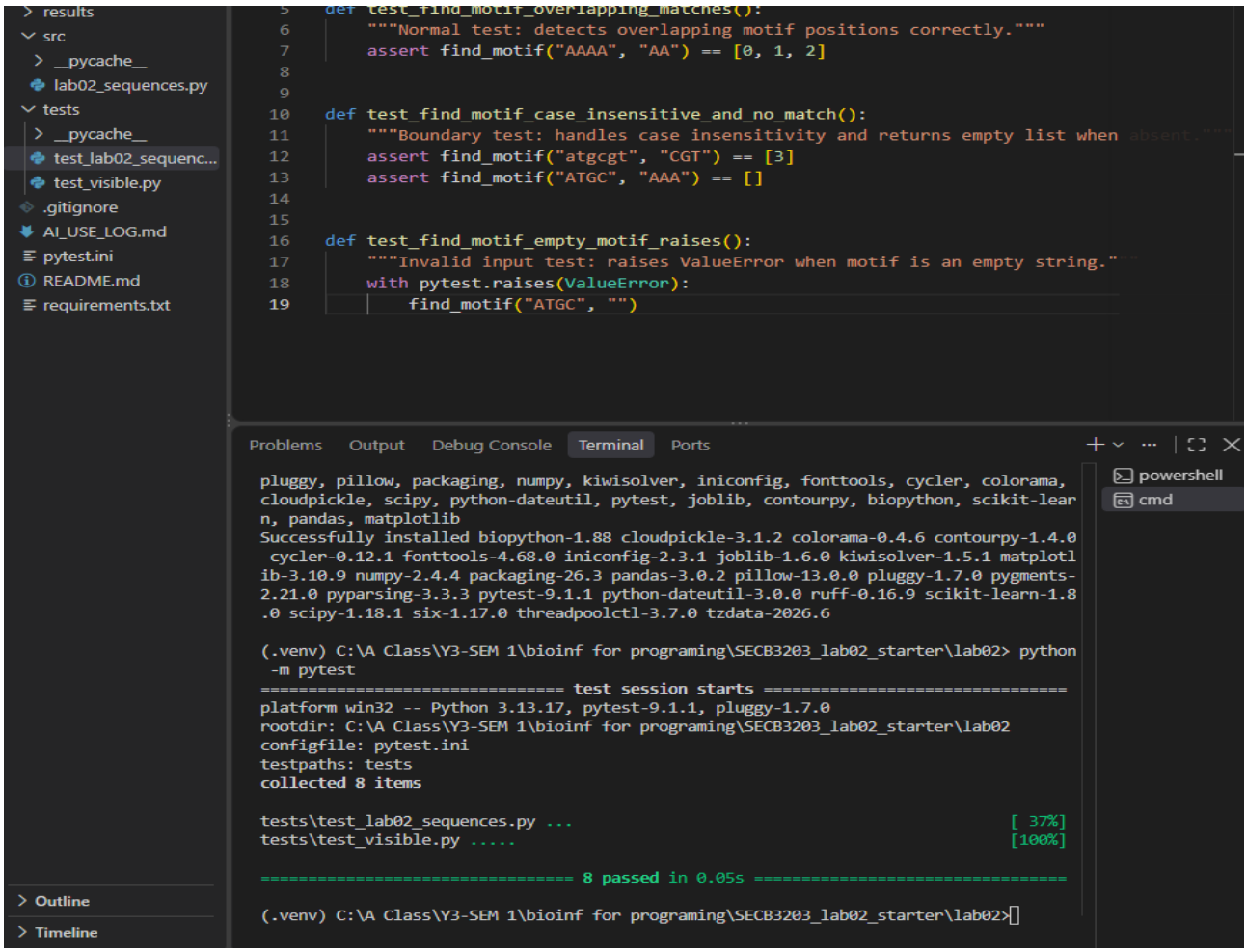
find_motif(seq, motif) returns every 0-based start position where motif occurs in seq, including overlapping matches, ignoring case. For example find_motif("AAAA", "AA") is [0, 1, 2]. Raise ValueError if motif is empty. Do not hard-code the motif length.

Hint: str.find(motif, start) accepts a start index. Move start forward by ONE base, not by len(motif).

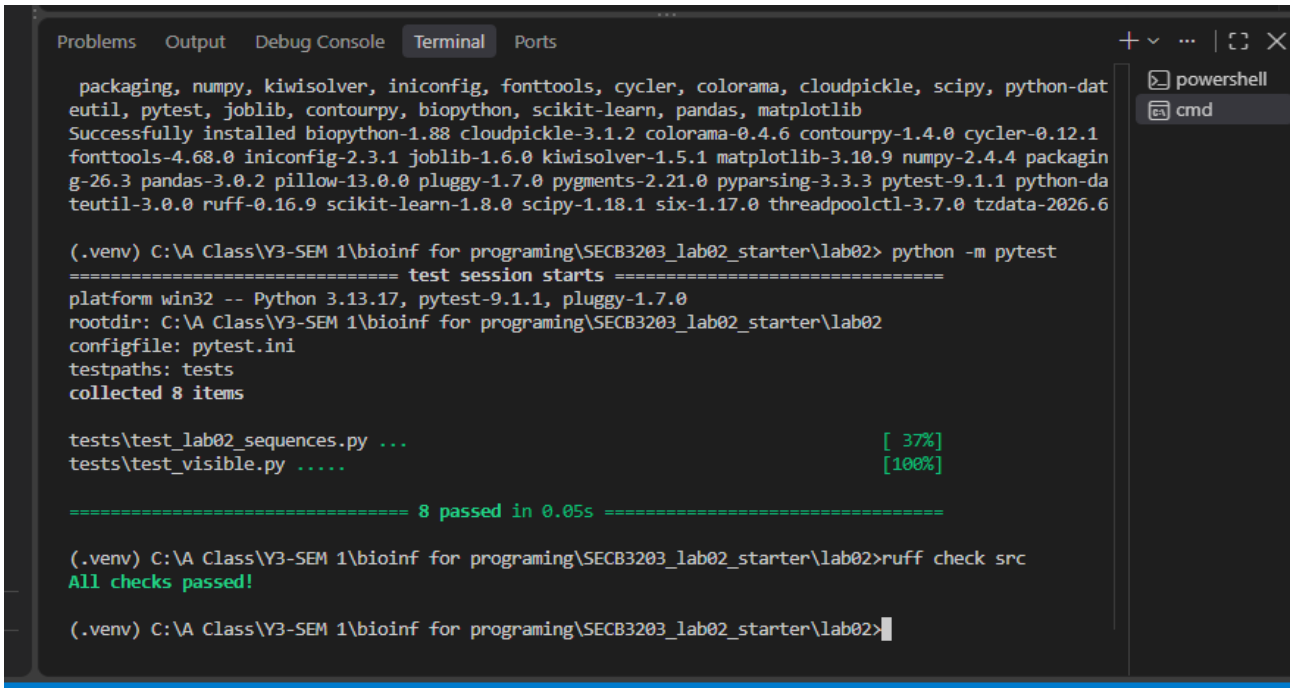
T5. Your own tests and clean code (4 marks)

Add three tests for find_motif, make `python -m pytest` pass, and make `ruff check src` report no issues. Complete your AI use log.

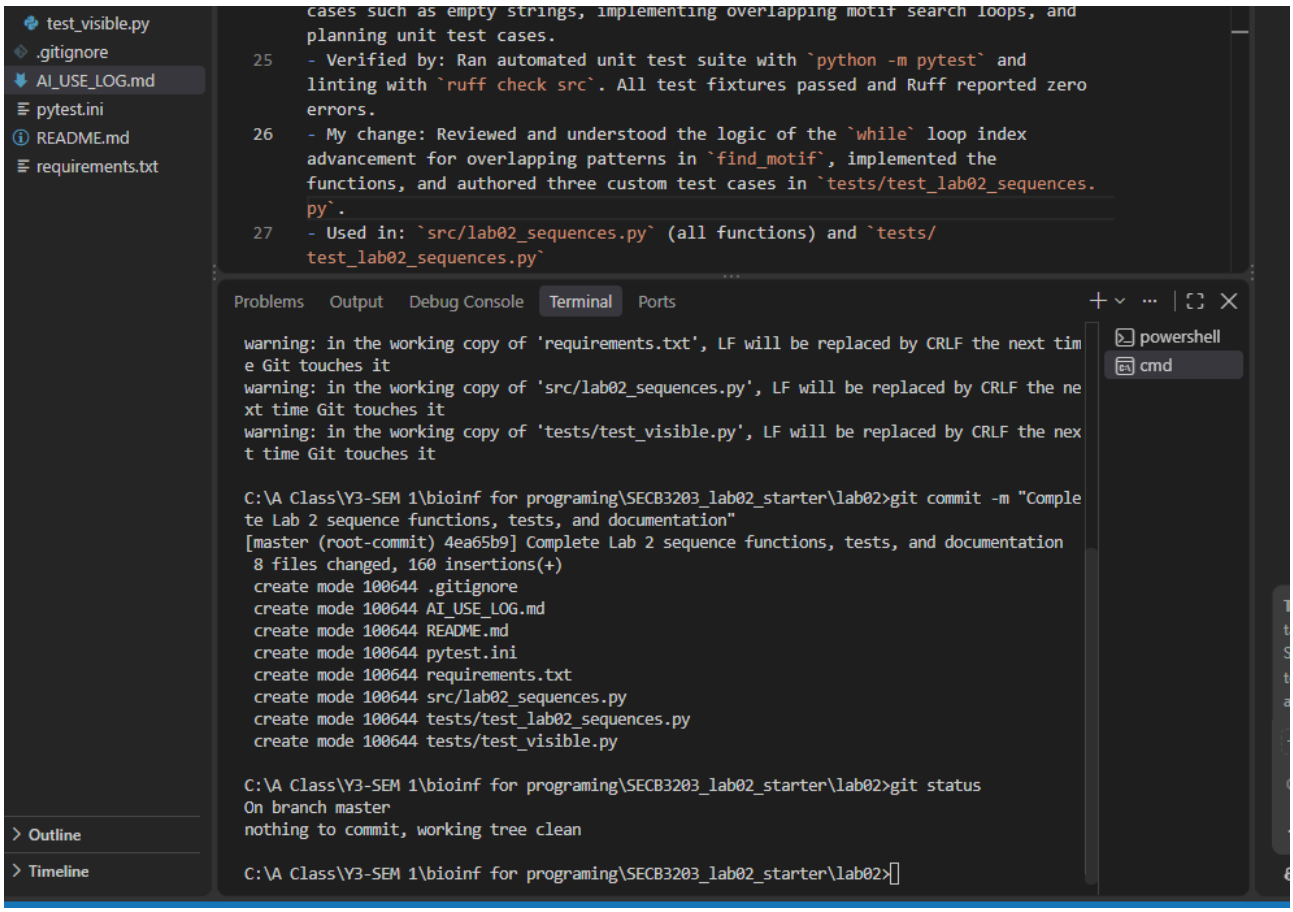
SCREENSHOT:



Terminal displaying green text showing all tests passed (both test_visible.py and test_lab02_sequences.py).



Task 5: Static code analysis with Ruff confirming zero PEP 8 style, naming, or formatting issues in src/.



Repository & Version Control: Git status output verifying clean working tree with all completed lab assets tracked and caches excluded.

What to submit

- Push your repository to GitHub (or submit the zip) before the lab deadline set in the course schedule.
- Include src/lab02_sequences.py, tests/test_lab02_sequences.py, AI_USE_LOG.md and a README.md with the exact commands to run the tests.

How it is marked

#	Criterion	Marks	Marked by
R1	clean_sequence and is_valid_dna correct (hidden tests)	3	Auto
R2	gc_content correct incl. lower case and invalid input	3	Auto
R3	transcribe correct incl. invalid input	2	Auto
R4	find_motif correct: overlaps, case, empty motif, no match	4	Auto
R5	Own tests present and passing (pytest exits 0 on the student test file)	2	Auto
R6	ruff reports no issues on src/	2	Auto
R7	AI use log submitted (prompt log or signed nil return) and valid	1	Auto
R8	Individual Q&A: explains one function, one test and one decision in own words	3	Lecturer
	Total	20	

Raw marks are scaled to 2% of the course mark.

Automated checks use hidden tests as well as the visible ones, so passing the visible tests alone does not guarantee full marks. You must be able to explain your own code in the individual Q&A.